higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

POWER MACHINES N5

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This marking guideline consists of 8 pages.

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NOTE: √ = ½ mark
✓ = 1 mark

QUESTION 1

1.1
- Oil firing
- Gas firing
- Coal/pulverised coal (3)

1.2
- The Edwards air pump
- The Hamilton pump OR rotary air pump OR rotating air pump (2)

1.3 The relationship between temperature and volume is directly proportional. (1)

1.4 Isochronism is the maintenance✓ of the equilibrium speed of a governor for every radius of ball rotation.✓

OR

A governor is isochronous if the equilibrium✓ speed is the same for every radius of ball rotation✓ (2)

Hunting means a governor cannot√ maintain√ a constant√ speed√. (2)
[10]

QUESTION 2

Adiabatic $P_1 =$ 150 kPa

m = 0,245 kg R = 0,293 kJ/kg.K

P_2 = 800 kPa Cp = 1,005 kJ/kg.K

Density = 1,2 kg/m^3

2.1
$$R = C_p - C_V V$$
$$C_V = C_p - R$$
$$C_V = 1{,}005 - 0{,}293$$
$$C_V = 0{,}712 kJ/kg.K$$
$$\gamma = \frac{C_p}{C_V}$$
$$\gamma = \frac{1{,}005}{0{,}712}$$
$$\gamma = 1{,}4$$ (3)

2.2

$$\rho = \frac{m}{V}$$

$$V_1 = \frac{m}{\rho}$$

$$V_1 = \frac{0,245}{1,2}$$

$$V_1 = 0,204m^3$$

$$P_1V_1 = mRT_1$$

$$T_1 = \frac{P_1V_1}{mR}$$

$$T_1 = \frac{150 \times 0,204}{0,245 \times 0,293}$$

$$T_1 = 426,273K$$ (6)

2.3

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}}$$

$$T_2 = T_1\left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}}$$

$$= (426.,273)\left(\frac{800}{150}\right)^{\frac{1.4-1}{1.4}}$$

$$= 687,704K$$ (4)

2.4

$$W = \frac{mR(T_1 - T_2)}{\gamma - 1}$$

$$W = \frac{0,245 \times 0,293(426,273 - 687,704)}{1,4 - 1}$$

$$W = -46,917\ kJ$$

OR **OR**

$$V_2 = \frac{mRT_2}{P_2}$$

$$V_2 = \frac{0,245 \times 0,293 \times 687,704}{800}$$

$$V_2 = 0,0617m^3$$

$$W = \frac{P_1V_1 - P_2V_2}{n - 1}$$

$$= \frac{(150 \times 0,204) - (800 \times 0,0617)}{1,4 - 1}$$

$$= -46,9\ kJ$$ (4)

2.5

$$\Delta U = -WD \quad \checkmark$$
$$= -(-46,917) \quad \checkmark$$
$$= 46,917\ kJ \quad \checkmark$$

OR **OR**

$$\Delta U = -WD$$
$$= -(-46,9) \quad \checkmark\checkmark$$
$$= 46,9\ kJ \quad \checkmark$$

OR **OR**

$$\Delta U = mC_V(T_1 - T_2)$$
$$\Delta U = 0,245 \times 0,712(687,704 - 426,273) \quad \surd\surd\surd$$
$$\therefore 45,6\ kJ \quad \checkmark$$

(3)
[20]

QUESTION 3

3.1

$$h_1 = 188\ kJ/kg$$
$$h_2 = h_f + xh_{fg}$$
$$\therefore 885 + (0,85 \times 1910)$$
$$\therefore 2508,5\ kJ/kg$$
$$h_3 = h_g + c_p(t_{ss} - t_s)$$
$$\therefore 2795 + 2,1(250 - 207)$$
$$\therefore 2885,3\ kJ/kg$$
$$EE = \frac{m_s(h_3 - h_1)}{m_f \times 2257}$$
$$\frac{m_s}{m_f} = \frac{EE \times 2257}{(h_3 - h_1)}$$
$$\therefore \frac{12 \times 2257}{(2885,3 - 188)}$$
$$\therefore 10,041 \frac{kg\ steam}{kg\ coal}$$

(7)

3.2
$$\eta = \frac{\dot{m}_s(h_3 - h_1)}{\dot{m}_f CV} \times 100$$
$$= \frac{10.041(2885.3 - 188)}{1 \times 32000} \times 100$$
$$= 84.67\%$$ (5)

3.3
$$heat\ transferred_{economiser} = \frac{m_s}{m_f}(h_2 - h_1)$$
$$= \frac{10{,}041}{1}(2508{,}5 - 188)$$
$$= 23300{,}141\ kJ/kg$$ (4)

3.4
$$heat\ transferred_{superheater} = \frac{m_s}{m_f}(h_3 - h_2)$$
$$= \frac{10{,}041}{1}(2885{,}3 - 2508{,}3)$$
$$= 3785{,}457\ kJ/kg$$ (4)

[20]

QUESTION 4

4.1
$$P = (B_m - M_m)\frac{101{,}325}{760}$$
$$= (760 - 670)\frac{101{,}325}{760}$$ ✓
$$= 11{,}999\ kPa$$ ✓
$$\approx 12\ kPa$$ ✓ (2)

4.2 $h_c = 179\ kJ/kg\ at\ 42,7^oC$

$At\ P = 12\ kPa$: ✓

$h_f = 207\ kJ/kg$

$h_{fg} = 2384\ kJ/kg$ ✓

heat emitted by steam = heat absorbed by water

√ $h_1\ at\ 12\ kPa = h_2$

$$100\left[207 + (0,95 \times 2384) - 179\right] = m \times 4,187 \times (40 - 21,1)$$

$$m = 2897,353\ \frac{kg}{\min} \times \frac{1\ \min}{60\ s} \quad \surd$$

$$m = 48,289\ \frac{kg}{s}$$

OR

Alternative Method:

$$h_c = c_p(t_s - t_c)$$

$$= 4,187(49,5 - 42,7)$$

$h_c = 28,472\ kJ/kg\ at\ 42,7^oC$

$At\ P = 12\ kPa$:

$h_{fg} = 2384\ kJ/kg$

heat emitted by steam = heat absorbed by water

$h_1\ at\ 12\ kPa = h_2$

$$100\left[(0,95 \times 2384) + 28,472\right] = m \times 4,187 \times (40 - 21,1)$$

$$m = 2897.949\ \frac{kg}{\min} \times \frac{1\ \min}{60\ s}$$

$$m = 48,299\ \frac{kg}{s}$$

(7)

4.3 $At\ t_c = 42{,}7^oC,\ P_s = 8{,}5\ kPa$

$P_T = P_a + P_s$
$P_a = 11{,}999 - 8{,}5$
$P_a = 3{,}499\ kPa$ ✓

$T_a = 273 + 38$
$T_a = 311\ K$ √

$P_aV_a = mRT_a$
$m_a = \frac{P_qV_a}{RT_a}$
$m_a = \frac{3{,}499 \times 2}{0{,}287 \times 311}$
$m_a = 0{,}078\ kg$ (5)

4.4 $P_aV_a = mRT_a$
$m_a = \frac{P_aV_a}{RT_a}$
$m_a = \frac{3{,}499 \times 2{,}5}{0{,}287 \times 311}$ ✓ √ √
$m_a = 0{,}098\ kg$ ✓ (4)

4.5 $\Delta m = 0{,}098 - 0{,}078$
$m_a = 0{,}02\ kg$ ✓ (2)
[20]

QUESTION 5

5.1 $U = \pi DN$
$= \pi \times 0.445 \times \frac{6500}{60}$
$= 151{,}45\ m/s$ (3)

5.2 See ANSWER SHEET (attached). (10)

5.3 5.3.1 $Cae = 105\ m/s$
5.3.2 $\varphi = 35^o$
(2 × 1) (2)
[15]

QUESTION 6

6.1

$$C_xH_y + \left(x+\frac{y}{4}\right)O_2 \rightarrow xCO_2 + \frac{y}{2}H_2O$$

$$C_4H_{10} + \left(4+\frac{10}{4}\right)O_2 \rightarrow 4\ CO_2 + 5\ H_2O$$

$$[(12\times4)+(10\times1)] + [(6.5\times2\times16)] \rightarrow [(4\times12)+(4\times2\times16)] + [(5\times2\times1)+(5\times16)]$$

$$58 + 208 \rightarrow 176 + 90$$

$$\frac{58}{58} + \frac{208}{58} \rightarrow \frac{176}{58} + \frac{90}{58}$$

$$1 + 3,586 \rightarrow 3,034 + 1,552$$

$$m = \frac{100}{23}[3,586]$$

$$= 15,591\ kg\ air/kg\ fuel$$

$$= 15,591 \times \frac{135}{100}$$

$$= 21,048\ kg\ air/kg\ fuel$$

(7)

6.2

Mass of the products:

$CO_2 = 3,034\ kg\ CO_2/kg\ fuel$

$H_2O = 1,552\ kg\ H_2O/kg\ fuel$

$O_2 = \frac{35}{100}\times 3,586$

$= 1,255\ kg\ O_2/kg\ fuel$

Or $O_2 = \left(\frac{23}{100}\times 21,048\right) - 3,586$

$= 1,255\ kg\ O_2/kg\ fuel$

$N_2 = \frac{77}{100}\times 21,048$

$= 16,207\ kg\ N_2/kg\ fuel$

$Total = 22,048\ kg\ products/kg\ fuel$

Percentages of the products:

$CO_2 = \frac{3,034}{22,048}\times 100 = 13.8\ \%$

$H_2O = \frac{1,552}{22,048}\times 100 = 7\ \%$

$O_2 = \frac{1,255}{22,048}\times 100 = 5,7\ \%$

$N_2 = \frac{16,207}{22,048}\times 100 = 73,5\ \%$

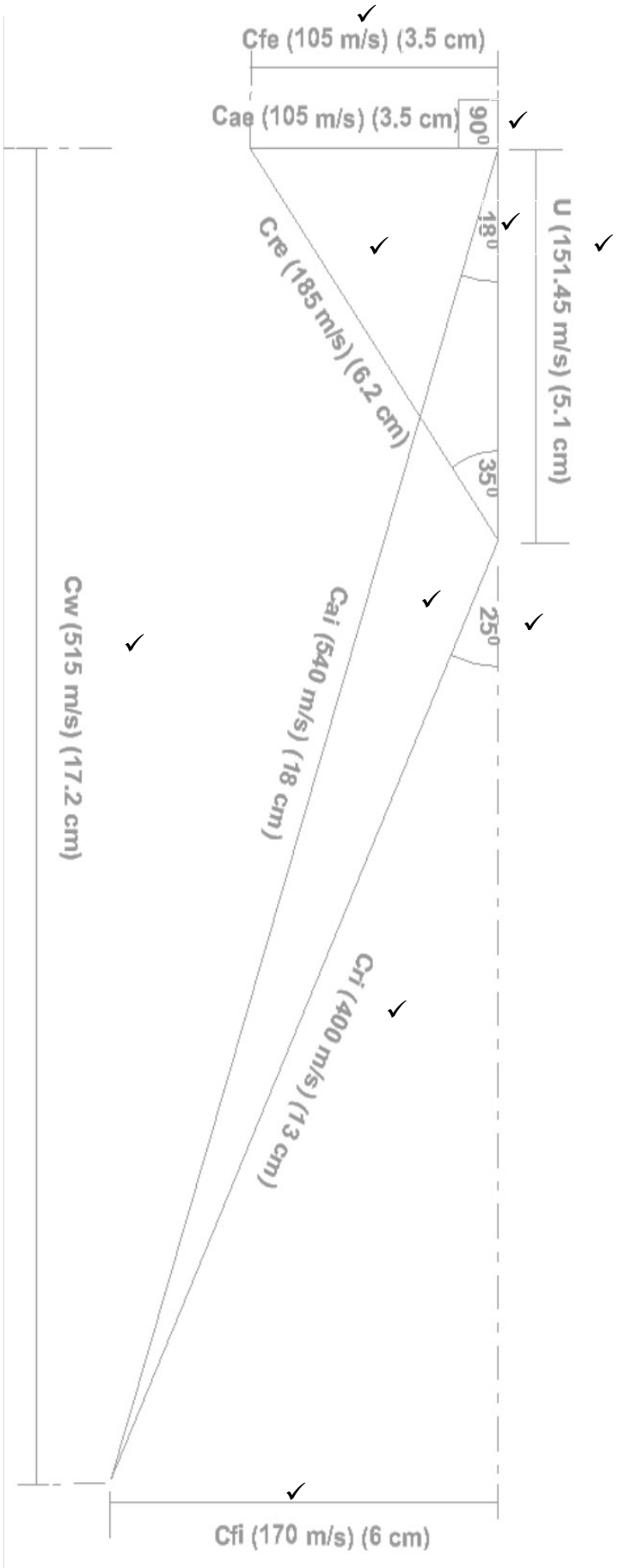
$Total = 100\ \%$

(8)

[15]

ANSWER SHEET

QUESTION 5.2



TOTAL: 100

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